

Machine Learning & Artificial Intelligence for Data Scientists: Welcome to the course

Ali Gooya

ali.gooya@glasgow.ac.uk

School of Computing Science

Who are we?

— — —



Dr Fani Deligianni
Senior Lecturer

Machine Learning
and Healthcare

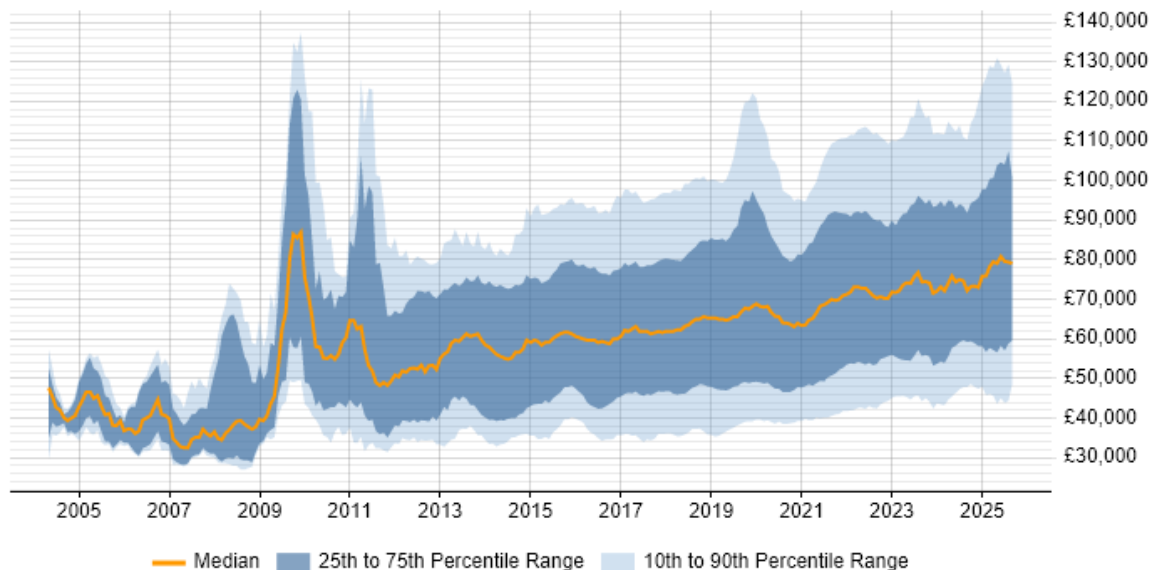


Dr Ali Gooya
Senior Lecturer

Machine Learning,
Computer Vision and
Medical Imaging

Why bother about machine learning?

Salary distribution trend for jobs in the UK citing Machine Learning.



What is Machine Learning?

— — —

Machine Learning (ML):

The field of study that gives computers the ability to learn without being explicitly programmed.

Arthur Lee Samuel was an American pioneer in the field of computer gaming and artificial intelligence. He popularized the term "machine learning" in 1959.



What is Machine Learning?

— — —

- Machine learning starts with **data**.
- **Data** is derived from **objects**:
 - Observations of people (preferences, health, etc)
 - Observations of the world (images, sounds, etc)

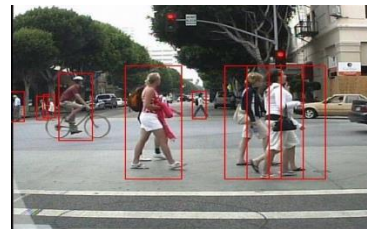
Examples of applied ML



Spam filtering



Face detection



Pedestrian detection



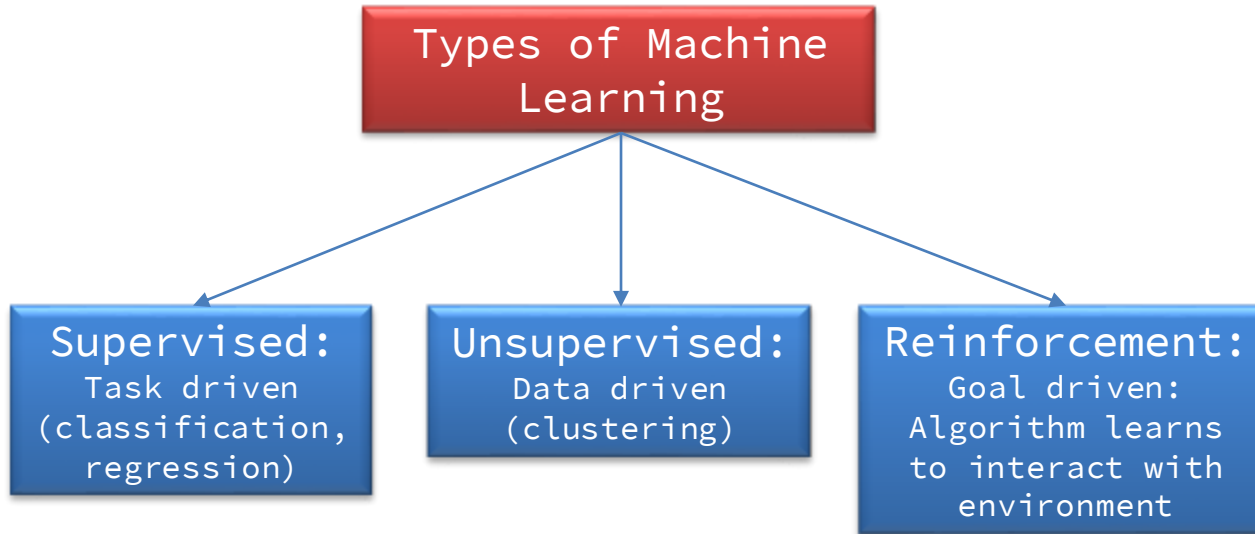
Movie recommendation



Voice recognition

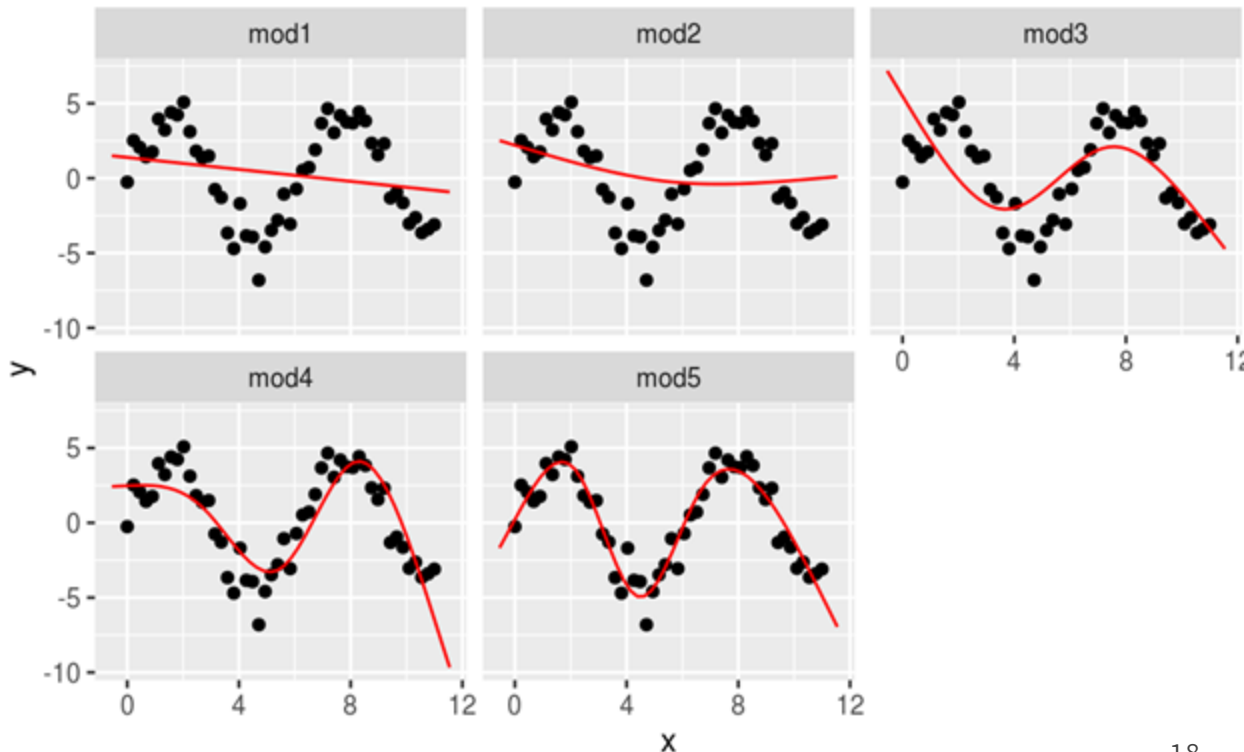
Types of Machine Learning

— — —



Supervised Learning: Regression

- Learning a continuous function from a set of examples.
- Example: Predicting stock prices (x might be time or some other variable of interest).



What do we want to achieve?

— — —

“All models are wrong, but some are useful.”

George E. P. Box

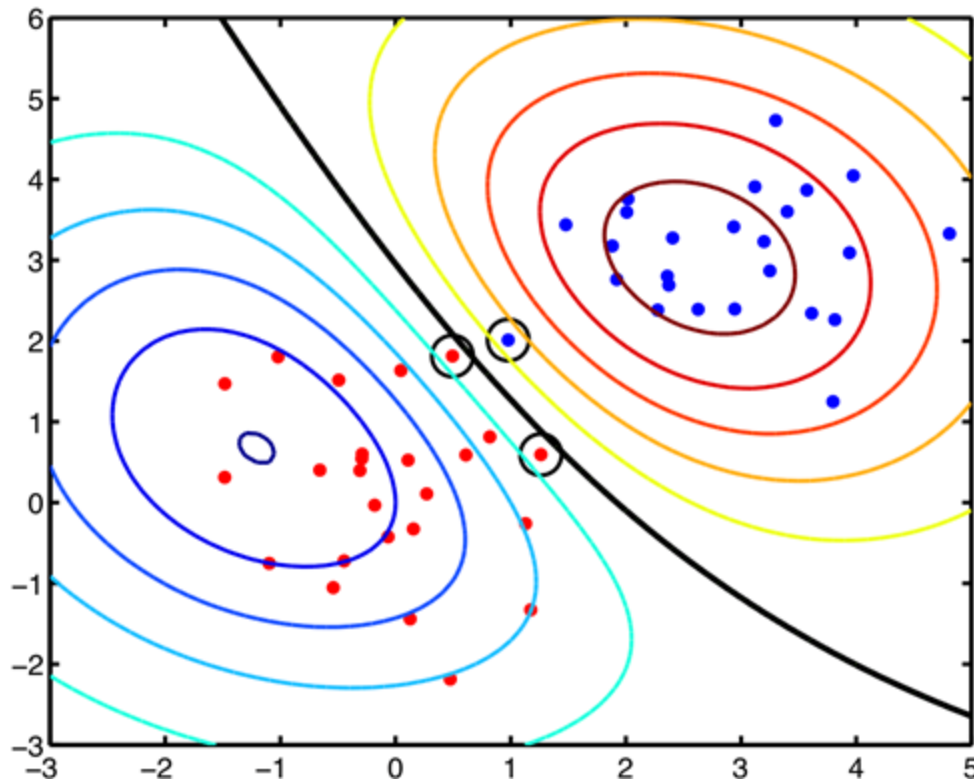
Now it would be very remarkable if any system existing in the real world could be exactly represented by any simple model. However, cunningly chosen parsimonious models often do provide remarkably useful approximations. For example, the law $PV = RT$ relating pressure P , volume V and temperature T of an “ideal” gas via a constant R is not exactly true for any real gas, but it frequently provides a useful approximation and furthermore its structure is informative since it springs from a physical view of the behavior of gas molecules.

For such a model there is no need to ask the question “Is the model true?”. If “truth” is to be the “whole truth” the answer must be “No”. The only question of interest is “Is the model illuminating and useful?”.



Supervised Learning: Classification

- Learning a rule that can separate objects of different types from one another
- Examples: Disease diagnosis, spam email detection.



Machine Learning Process

Data Collection

Feature Selection

Model choice

Training

Evaluation

Most of this
class

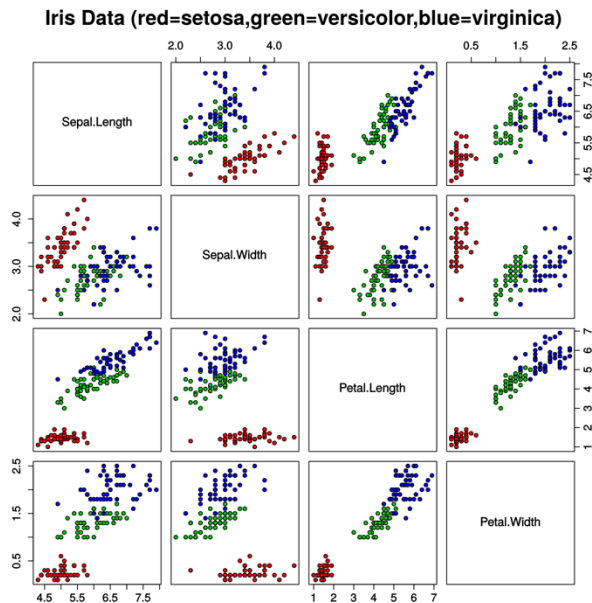
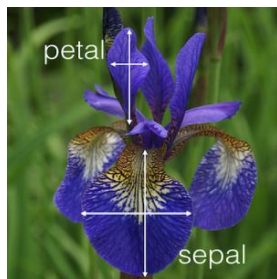


Machine Learning Process

Data Collection:
For example, many
images of various
iris flowers

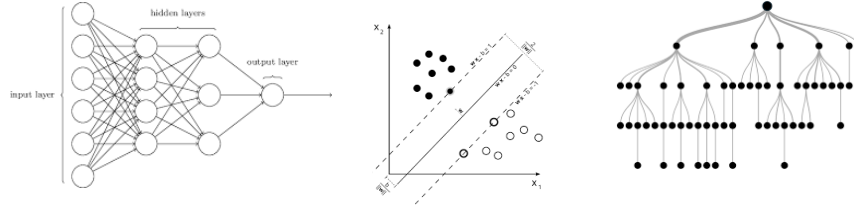


Feature Selection:
E.g., length/widths
of petal and sepal in
each flower

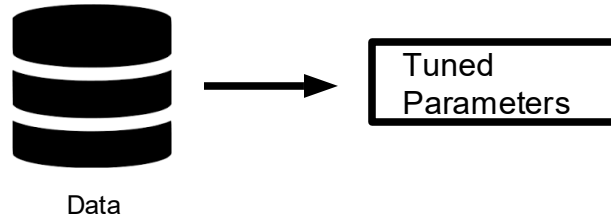


Machine Learning Process

Model choice



Training

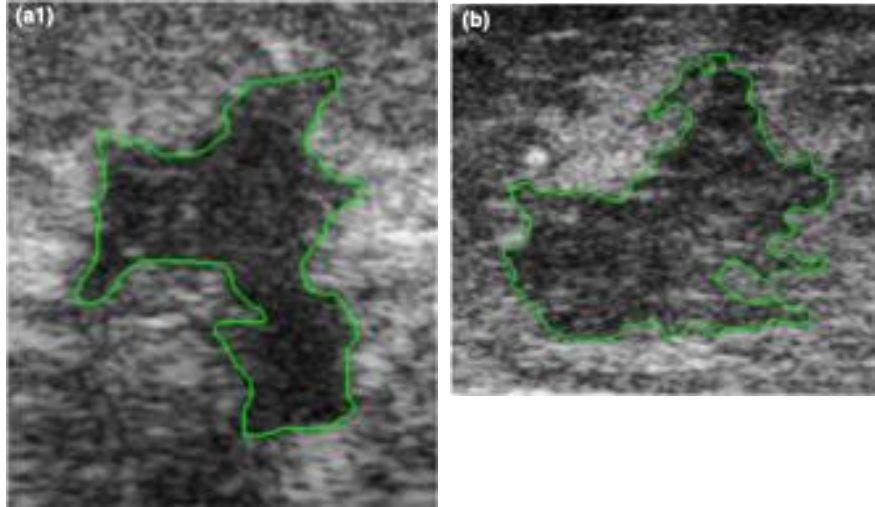


Evaluation

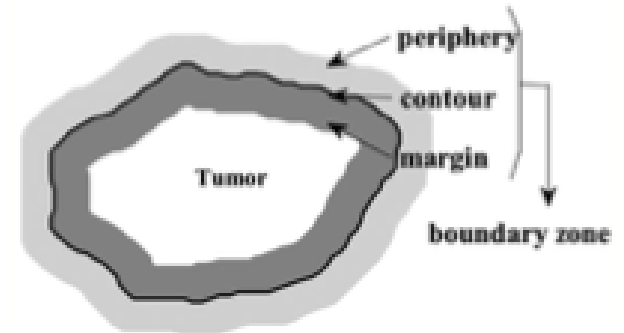
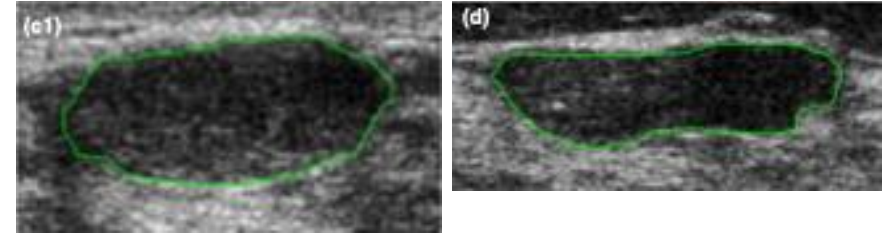
		Predicted Class		
		Setosa	Versicolor	Virginica
Actual Class	Setosa	14	1	1
	Versicolor	1	11	3
	Virginica	1	3	10

Classification of breast lesions in ultrasonography using sparse logistic regression and morphology-based texture features, Medical Physics, 2019

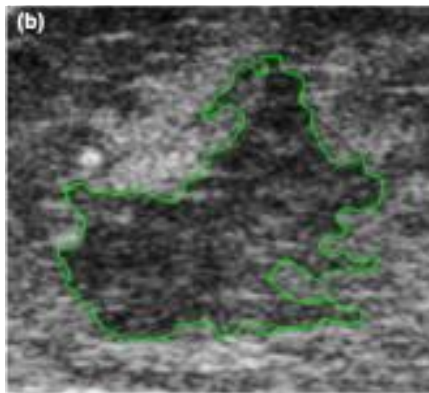
Samples of malignant tumours



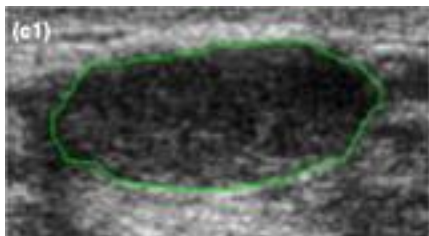
Samples of benign tumours



Classification of breast lesions in ultrasonography using sparse logistic regression and morphology-based texture features, Medical Physics, 2019



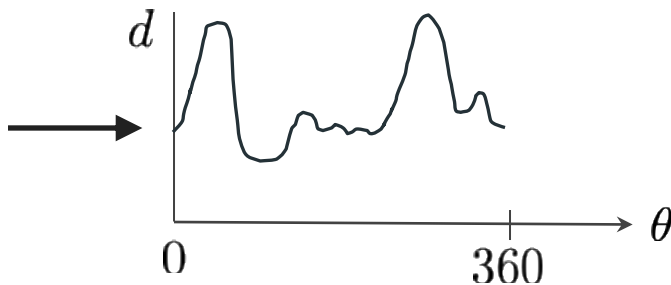
Malignant



Benign

Classification of breast lesions in ultrasonography using sparse logistic regression and morphology-based texture features, Medical Physics, 2019

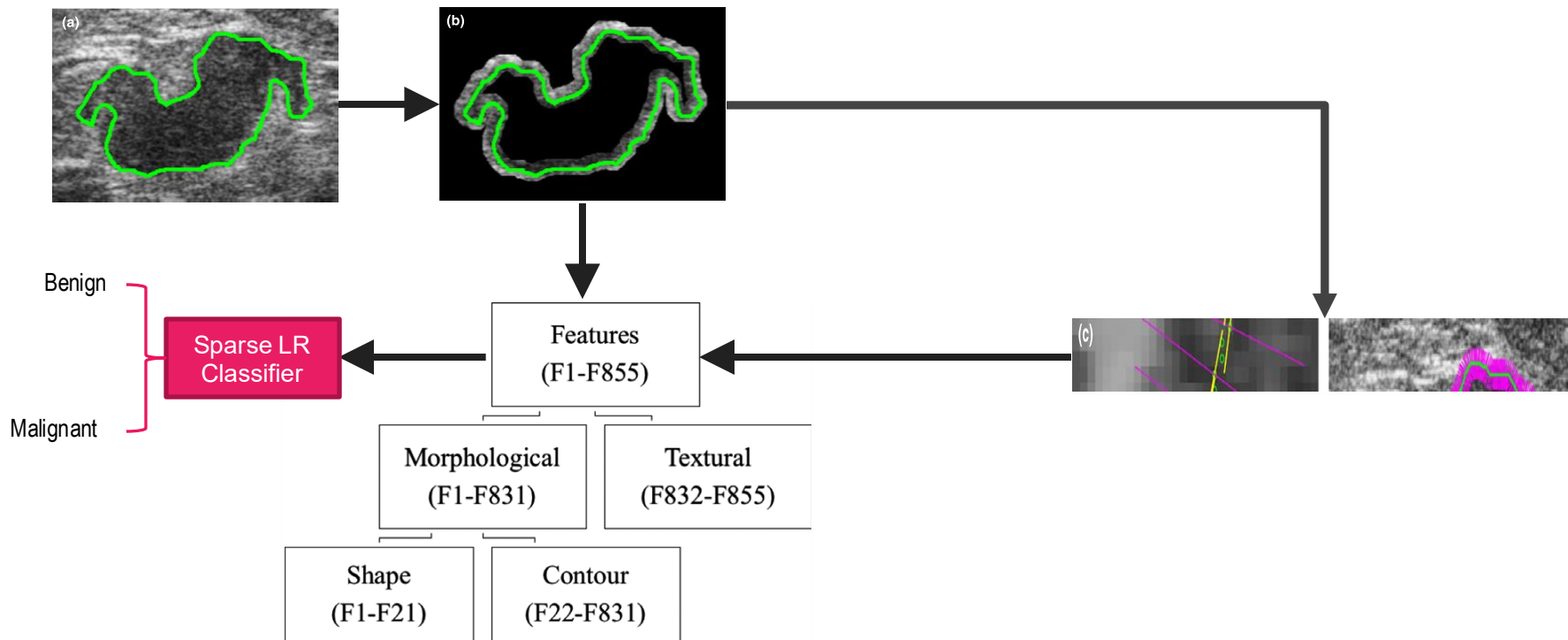
Extracting contour-based features



Fourier descriptors

Classification of breast lesions in ultrasonography using sparse logistic regression and morphology-based texture features, Medical Physics, 2019

- Applied to 104 breast US images of tumours, including 72 benign and 32 malignant

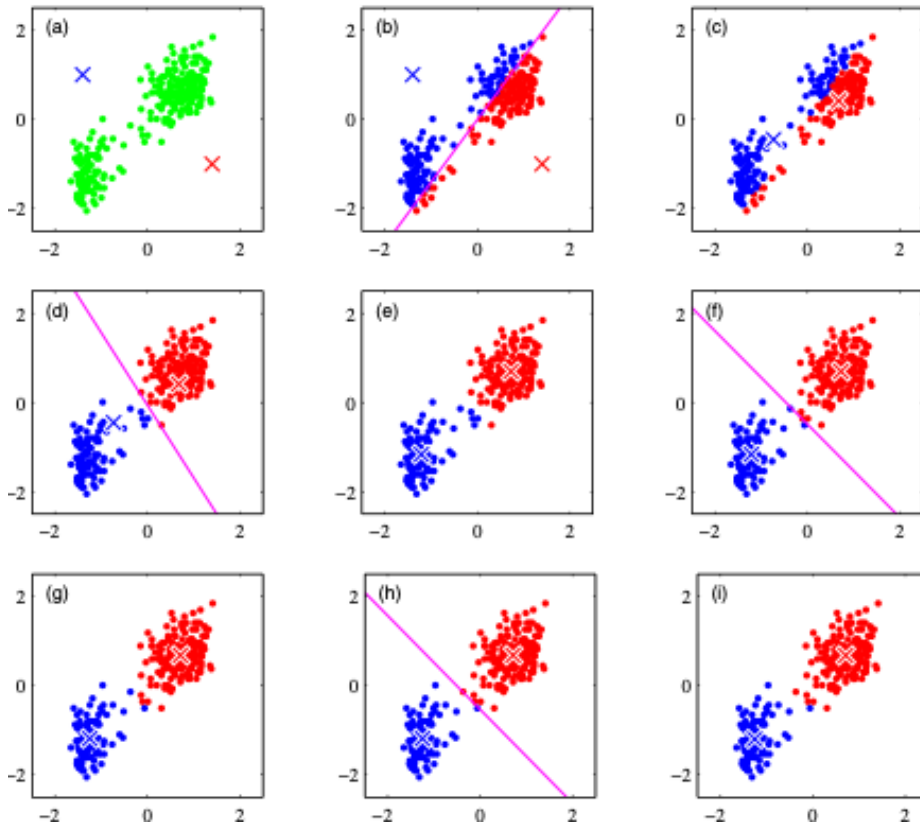


Unsupervised Learning: Clustering

- Finding groups of similar data points.
- Example: clustering re-scaled Old Faithful dataset



Observation	Eruption Duration (min)	Waiting Time (min)
1	3.6	79
2	1.8	54
3	3.333	74
4	2.283	62
5	4.533	85
6	2.883	55
7	4.7	88
8	3.6	85
9	1.95	51
10	4.35	85
11	1.833	54
12	3.917	84
13	4.2	78
14	1.75	47



Clustering (color quantisation)

— — —

$K = 2$



$K = 3$



$K = 10$

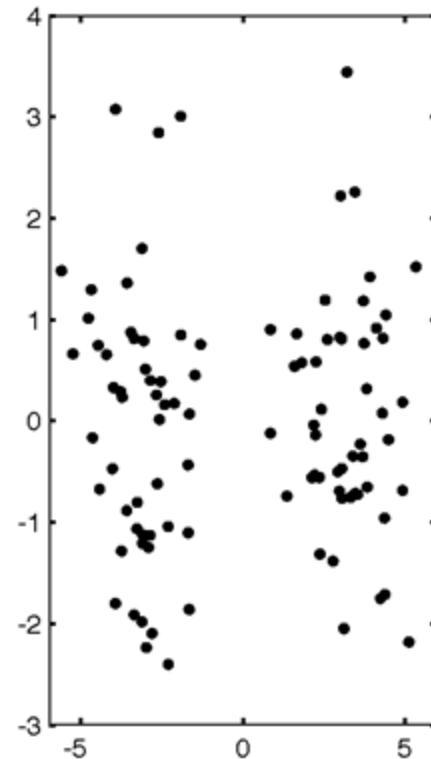
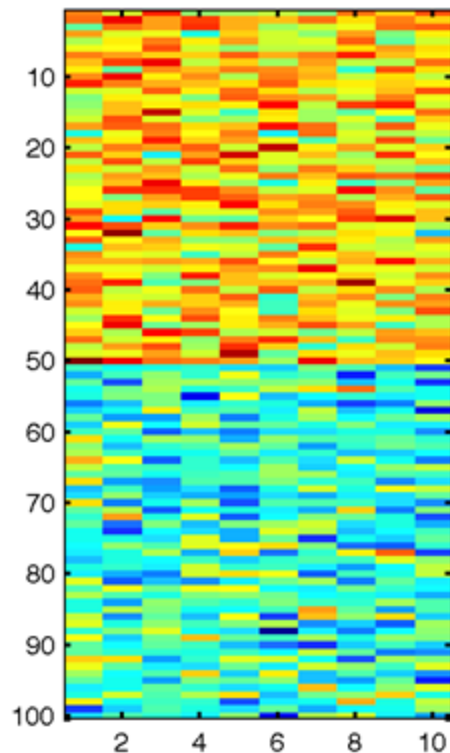


Original image



Projection (Dimensionality Reduction)

- Reducing the number of variables - e.g. from 10 to 2.
- Visualising complex data.



Course structure

- Topics covered
 - What is “learning from data”
 - Introduction to the problems of Regression, Classification, Clustering, and Projection.
 - Training, validation, and testing.
 - Performance metrics and the importance of robust baselines.
 - Common pitfalls.
 - Presentation of results.

Course structure

— — —

- **3 units of case studies** covering different datasets, visualisation techniques, models and learning algorithms. Taught by multiple staff.

Case study

- A dataset and predictive / exploratory problem to be solved.
- An introduction to one or more algorithms.
- A practical session (1 hr in lab plus 1hr in students' own time).
- A wrap-up session .

Assessment and feedback

— — —

- Exam: 60%, topics covered in lectures.
- Coursework: 40%, 2 case studies.
- Written feedback for the coursework.
- 1-1 help from GTAs and demonstrators in the labs.
- Lectures where solutions or common issues are discussed.

-- --

Don't panic! Have fun!